Can Photos Prove an Aero-Engine Blade Is Damaged?

Finding 3D damage from a CAD model and a few photos, and knowing when the photos are not enough

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1 Introduction: why this matters

Aero engines (jet engines) contain hundreds of compressor and turbine blades. Small damage such as a **dent**, a **nick** on the edge or a **crack** can grow and lead to costly failures, so blades are inspected regularly [1]. Today this is mostly done by people looking at photos or videos, which is slow and depends on the inspector. But every blade is made from a known design, its **CAD model**, so we know exactly what a healthy blade should look like. This research compares photos of a blade with its healthy CAD model to find, in 3D, **where** the blade is damaged and **how large** the damage is.

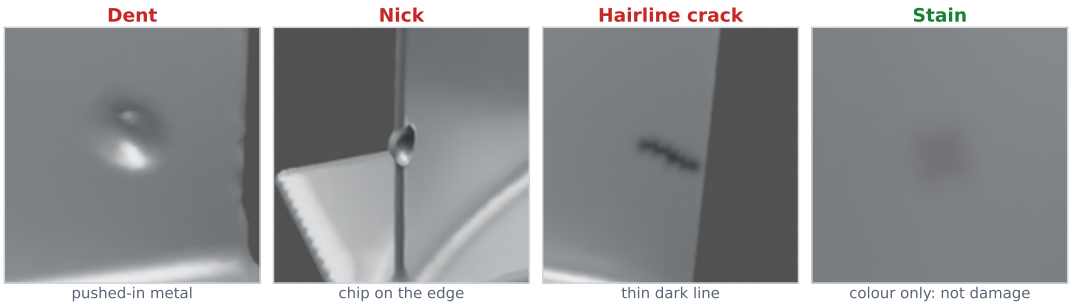


Figure 1. The damage types studied (computer-rendered close-ups). A stain changes only the colour and is not damage.

2 Problem statement

A dark spot in a photo can have **four different causes** (Figure 2). Only one of them is real damage, yet in a photo they can look almost the same: a 0.1 mm dent changes the image only slightly, while moving the camera by 1° changes thousands of pixels. A computer that simply looks for "something different" raises many **false alarms**, and one that always gives an answer will sometimes **invent damage** that is not there.

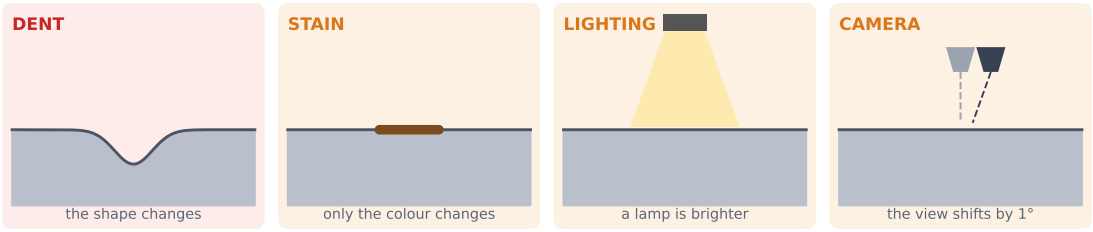


Figure 2. One spot, four possible causes. Only the first is physical damage.

Research questions. (1) Using a CAD model and a few photos, can we tell real damage apart from stains, lighting changes and camera errors? (2) Can we measure the damage in 3D, i.e. its location and depth in mm? (3) Can the method recognise *when the photos are not enough* and say so, instead of guessing?

3 Objectives

Main goal: a method that reports blade damage in 3D *only when the photos can prove it*, and otherwise says "not enough evidence".

1. Separate real damage from stains, lighting and camera errors.
2. Locate the damage on the 3D blade and estimate its depth.
3. Keep false alarms low and controlled (about 5%).
4. Recognise the type of damage: dent, nick or crack.
5. Test on computer-rendered blades, then on real 3D-printed, damaged and scanned blades.

Benefit: faster, more consistent inspections that an engineer can trust, because the method states how sure it is.

4 Preliminary literature review

Recent methods compare photos with a model of a healthy object and mark *unusual pixels*, even when the camera or lighting changes [2,3]. Other methods use a CAD model to build a 3D copy of an object from a few photos [4]; there, differences from the CAD are treated as noise, not as damage. 3D anomaly datasets give anomaly scores, not damage sizes [5]. Our core tool, "render and compare", builds on differentiable rendering [6].

Existing work	Marks damage in 2D	Measures size in 3D (mm)	Says "not sure"
Anomaly detection [2,3]	✓	×	×
CAD-based 3D reconstruction [4]	×	×	×
3D anomaly datasets [5]	✓	×	×
This research	✓	✓	✓

Figure 3. What existing work provides. **The gap:** no method measures damage in 3D *and* says when the photos cannot support a conclusion. An author of [3] independently suggested this question as a valuable research contribution.

5 Methodology

Main idea, "render and compare". The computer draws (renders) the healthy blade from its CAD model and compares the drawing with the photos, in three steps (Figure 4). "Clearly better" is a threshold set on healthy and stained blades, so only about 5% of undamaged blades are flagged. To recognise the damage type, separate *specialists* for dent, nick, crack and stain each try to explain the photos, and the best one wins.

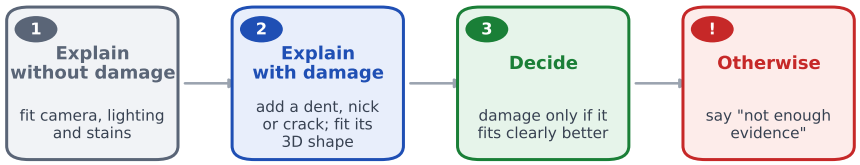


Figure 4. The method in three steps.

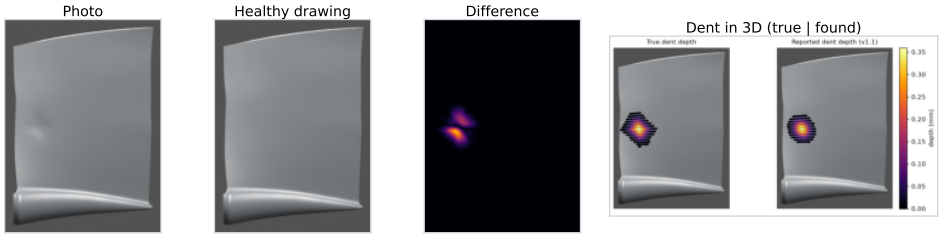


Figure 5. A worked example. The difference between the photo and the healthy drawing shows where to look; the dent found in 3D is 0.35 mm deep (true depth: 0.36 mm).

Test data. Over 1,300 computer-rendered cases with exactly known damage, on public compressor blades and six industrial blade designs. The test blades were kept hidden until the method was final.

6 Preliminary results

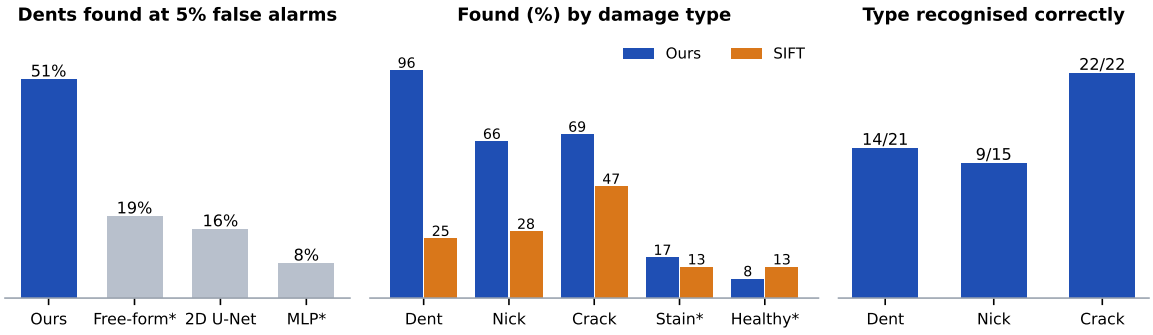


Figure 6. Left: dents found when every method is allowed the same 5% false alarms (*our versions of the CAD-based method [4]). Middle: damage found on new blade designs, ours vs. a standard image-matching method (SIFT); *stains and healthy blades should *not* be reported. Right: damage type recognised correctly, for damage found in the right place.

- **Finds damage reliably:** 51% of visible dents at 5% false alarms (other methods: at most 19%); 12 of 13 dents deeper than 0.2 mm; sharp nicks 66% and hairline cracks 69% on new designs.
- **Works on new blade designs:** false alarms dropped from 51% to 4% after one fix.
- **Recognises the damage type:** with improved nick and crack specialists, the type is correct for all 22 cracks and 9 of 15 nicks (for 6 more nicks it honestly says "type uncertain"), and the winning specialist stays in the right place in 42 of 43 cases.
- **Calibration is necessary:** textbook statistical tests give 32% and 96% false alarms instead of the intended 1%.
- **Main limitation:** the location is reliable, but on new designs depth is measured about 40% too shallow; more cameras do not fix this, so better lighting is the next step.

7 Next step: real blades

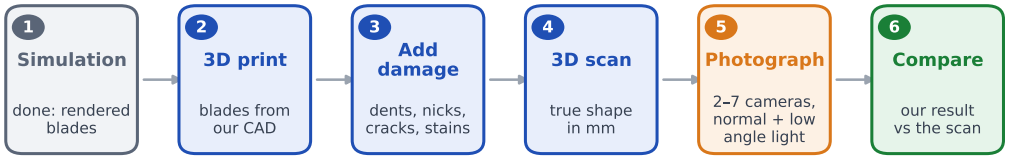


Figure 7. From computer-rendered blades to real ones: we compare our measurements with the 3D scan, using the same measures as above.

References

[1] M. A. Mohammed Eltoum et al., BladeSynth: a rendering-based synthetic dataset for aero-engine blade defect inspection, *Scientific Data*, 2025. [2] Zhou et al., PAD: a dataset and benchmark for pose-agnostic anomaly detection, 2023. [3] Yang et al., PIAD: pose and illumination agnostic anomaly detection, 2025. [4] Overdulve et al., CADSplat: sparse-view Gaussian splatting aided by CAD models, 2026. [5] Costanzino et al., SiM3D: single-instance multiview multimodal 3D anomaly detection benchmark, 2025. [6] Laine et al., Modular primitives for high-performance

differentiable rendering, 2020.